

Introduction to Knot Theory

(2) Knot Group and Wirtinger Presentation

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The Knot Complement

Idea

A knot is embedded in space.

Instead of studying the curve itself, we study the **space around the knot**.

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- Intuition: how the knot twists the surrounding space
- Many knot invariants come from the topology of this complement

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Conclusion

If two knots are equivalent, their complements are homeomorphic.

Thus topology of the complement gives knot invariants.

Knot Group

Definition

The **knot group** of a knot K is

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Remark

We may also work in $\mathbb{R}^3$ instead of S^3 .

$$\pi_1(S^3 \setminus K) \cong \pi_1(\mathbb{R}^3 \setminus K)$$

Wirtinger Presentation

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Procedure

- ① Assign a generator to each arc of the diagram
- ② For each crossing, write a relation
- ③ Obtain a group presentation

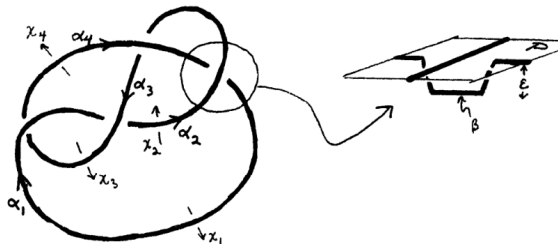
Wirtinger Presentation

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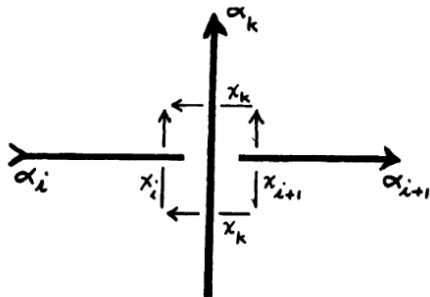
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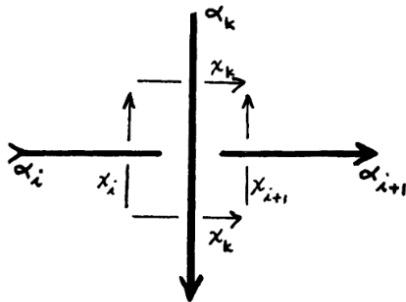


Wirtinger Relation

At each crossing we obtain a relation.



$$r_1 : \quad x_k x_i = x_{i+1} x_k$$



$$x_i x_k = x_k x_{i+1}$$

Finite Presentation

Theorem

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Therefore

$$G(K) = \langle \text{arcs} \mid \text{crossing relations} \rangle$$

Trefoil Knot Example

For the trefoil diagram we obtain

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This group is also equivalent to

$$G(K) = \langle a, b \mid a^2 = b^3 \rangle$$

Reidemeister Moves and Wirtinger

If two diagrams are related by Reidemeister moves

$$D_1 \leftrightarrow D_2$$

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Idea

Reidemeister moves introduce generators and relations that cancel algebraically.

Thus the knot group is a knot invariant.

First Homology

Theorem

For every knot K

$$H_1(S^3 \setminus K) \cong \mathbb{Z}$$

The first homology group cannot give information

Alexander Duality

Alexander duality states

$$\tilde{H}_i(S^3 \setminus K) \cong \tilde{H}^{2-i}(K)$$

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Since

$$K \cong S^1$$

we obtain

$$H^1(S^3 \setminus K) \cong \mathbb{Z}$$

Tubular Neighborhood

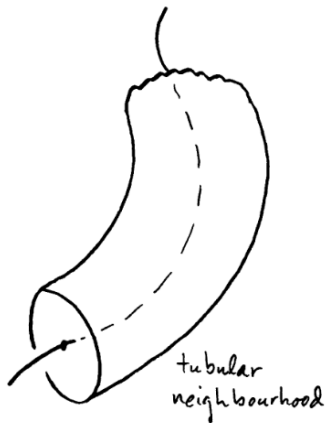
A knot has a neighborhood homeomorphic to

$$V(K) \cong S^1 \times D^2$$

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Boundary Torus

The boundary of the neighborhood is

$$\partial V(K) \cong T^2$$

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Two important curves appear on this torus.

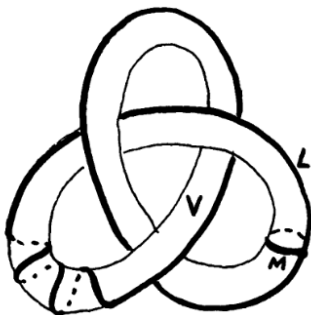
- Meridian M
- Longitude L

Meridian and Longitude

- Meridian M : boundary of the disk D^2
- Longitude L : curve along the knot

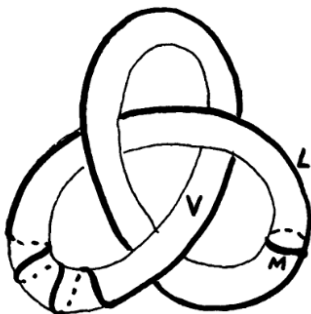
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The longitude will be defined using a **Seifert surface** (Week 4).

Peripheral Subgroup

The torus boundary induces a subgroup

$$\langle M, L \rangle \subset \pi_1(S^3 \setminus K)$$

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This subgroup corresponds to

$$\langle M, L \rangle \cong \pi_1(T^2) \cong \mathbb{Z}^2$$

Why Peripheral Structure Matters

The knot group alone does not determine the knot.

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There exist knots with

$$G(K_1) \cong G(K_2), \text{ but } K_1 \neq K_2$$

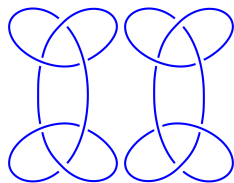


Figure: Granny knot and square knot

https://en.wikipedia.org/wiki/Peripheral_subgroup

Why Peripheral Structure Matters

Gordon–Luecke Theorem claims that a knot is determined by its complement.

Gordon–Luecke Theorem

For knots K, K' , if there exists a orientation-preserving homeomorphism between complement space,

$$S^3 \setminus K \cong S^3 \setminus K'$$

two knots are equivalent.

Why Peripheral Structure Matters

Let the peripheral subgroup $P(K) = \langle M, L \rangle$ and $G(K) = \pi_1(S^3 \setminus K)$.
 $(G(K) = \pi_1(S^3 \setminus K), P(K) = \pi_1(\partial(V(K))))$ determines the manifold $(S^3 \setminus K)$.

Corollary

For knots K, K' , if there exists a (S^3) orientation preserving homeomorphism s.t.

$$(G(K), P(K)) \cong_{S^3} (G(K'), P(K'))$$

two knots are equivalent.

Abelianization

If we abelianize the knot group

$$h : G(K) \rightarrow G(K)^{ab} = G(K)/[G(K), G(K)]$$

we can obtain a Hurewicz map $h : \pi_1(S^3 \setminus K) \rightarrow H_1(S^3 \setminus K)$.

In the abelianization, all commutators vanish., so

$$G(K)^{ab} \cong H_1(S^3 \setminus K)$$

Consequence

Since

$$H_1(S^3 \setminus K) \cong \mathbb{Z}$$

we obtain

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Conclusion

Abelianization loses almost all information about the knot.

Therefore we need stronger invariants.

References

- D. Rolfsen, *Knots and Links*, AMS Chelsea.
- A. Hatcher, *Algebraic Topology*, Cambridge Univ. Press.
- C. McA. Gordon and J. Luecke, *Knots are determined by their complements*, J. Amer. Math. Soc. 2 (1989), 371–415.
- F. Waldhausen, *On irreducible 3-manifolds which are sufficiently large*, Annals of Mathematics, Second Series 87 (1968), no. 1, pp. 56–88. DOI: 10.2307/1970594

Images

- Rolfsen, *Knots and Links*
- Wikipedia (Peripheral subgroup)